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CLINICAL RESEARCH

Validation of bioelectrical impedance analysis in the evaluation of body composition in patients with breast cancer

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Abstract

Background: The evaluation of body composition is an essential parameter for preventing obesity and sarcopenic obesity, which are prognostic factors in breast cancer. This study aims to validate the bioelectrical impedance analysis (BIA) of women who are breast cancer survivors by using the dual-energy x-ray absorptiometry (DXA) measurement method.

Methods: This validation study included 104 women without metastasis between 32 and 72 years old (mean 47.03 ± 8.59) whose treatment was completed 6 months prior. Body composition analysis was performed sequentially using both measurements and when participants were hungry.

Results: Meaningful differences were found in fat-free mass (FFM) (BIA: 46.57 ± 5.54 kg; DXA: 41.06 ± 5.11 kg), body fat percentage (%BF) (BIA: $34.28\% \pm 6.24\%$; DXA: $43.91\% \pm 5.58\%$), body fat mass (FM) (BIA: 25.37 ± 8.84 kg; DXA: 31.24 ± 9.09 kg), and lean soft tissue mass (LSTM) (BIA: 4.42 ± 5.66 kg; DXA: 38.75 ± 4.98 kg) ($P < 0.001$). Powerful associations for body FM and strong associations for other parameters were seen. A constant and/or proportional error was found between the two devices within the direction of strong and solid components. Compared with DXA, the BIA measurement gives a lower estimate of %BF and FM and a higher estimate of LSTM and FFM.

Conclusions: By the mathematical relationship between the two measurement methods, it seems possible to adapt the body composition parameters obtained from BIA of patients with breast cancer to DXA results. In the future, there will be a need to evaluate these two devices with more extensive studies.

KEYWORDS
bioelectrical impedance analysis, body composition, breast cancer, dual-energy x-ray absorptiometry, sarcopenia

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INTRODUCTION

Bioelectrical impedance analysis (BIA) is one of the most common methods used in clinical practice to measure body composition.[1] It is clear that some diseases can affect patients' body composition, but this method is not designed according to different disease populations.[1]

Breast cancer represents one in four cancer diagnoses in women. In the Global Cancer Observatory (GLOBOCAN) 2020 report, breast cancer was, for the first time, reported as the most commonly diagnosed cancer for both sexes and in the women's group (11.7% for both sexes and all age groups; 24.5% for women and all age groups).[2] In addition to traditional prognostic factors (tumor size, lymph node involvement, tumor histology, etc), new clinical studies and biological markers have become the focus of studies on improving the management of this disease.[2–5] Obesity is being studied as an independent prognostic factor that adversely affects the prognosis in breast cancer.[6] During the treatment and the follow-up timeline of breast cancer, it is recommended to carefully monitor the change(s) in the patient's body composition and determine the patient's necessary lifestyle changes. In addition to oncological treatment, evaluation and monitoring of the patient's nutrition status are known to be the hallmarks of good clinical practice in oncology.[2–7] The evaluation of the body composition grants knowledge regarding both the nutrition status and the functional capacity of the human body, and body composition is also used in evaluating the results of treatment interventions, designing nutrition strategies, making assessments tied to the disease and health, and identifying growth and development in all ages.[8–10]

Sarcopenia is a condition defined as a syndrome associated with loss of muscle mass and strength and decreased physical performance in older adults. Many complex mechanisms lead to sarcopenia, including neurodegeneration, impaired signaling, inflammation, disuse, and reduced nutrient intake. Sarcopenia is prevalent in adults with cancer due to the increasing prevalence of disease with age. Furthermore, inflammation and malnutrition associated with cancer may worsen muscle. Many studies have reported that the presence of sarcopenia in patients with cancer is associated with adverse clinical outcomes, such as postoperative complications; and in the past decade, a notable number of studies have reported an association between sarcopenia and mortality rate among women with breast cancer. Certainly, mortality is one of the most important clinical outcomes in clinical oncology.[11]

Although the underlying causes of weight gain after breast cancer diagnosis are not fully understood, it are associated with increased cancer recurrence and decreased survival. It is important to note that in women undergoing breast cancer treatment, fat mass (FM) increases more than fat-free body mass, and thus, sarcopenic obesity develops.[7]

It is not possible to capture this change in body composition using body mass index (BMI; defined as weight divided by square height) measurements. More sophisticated and precise methodologies are required to decipher the relationships between body composition and breast cancer outcomes. As a noninvasive, widely used, and inexpensive body composition measurement method, the validation of BIA regarding the evaluation of the body composition of patients with breast cancer may shed light on further need of body composition studies to be conducted in this field.

The most frequently used methods in clinical practice and epidemiology studies for body composition evaluation are BIA and dual-energy x-ray absorptiometry (DXA). BIA has been favored in recent years because it is both an inexpensive and noninvasive method. By contrast, DXA is a 3-Compartment (3C) model measurement method that can measure the entire body composition. The whole body can be scanned to measure bone mass and soft tissue composition. However, DXA is a method not used in routine clinical practice for body composition in this patient group because the patient undergoes low-dose radiation exposure during measurement. Also, DXA testing is expensive.[7,12]

Body composition calculation with BIA measurement is based on the formulization of measurements obtained from healthy individuals. One limitation of using BIA in this populations is that the body composition of patients with breast cancer may be different from that of healthy individuals.

In this study, the aim was to validate the usage of BIA (an inexpensive, noninvasive, and practical body composition method) and DXA (considered to be the gold-standard method for evaluating the body composition of patients with breast cancer), and to determine safe and correct results of BIA measurement for patients with breast cancer by the use of coefficients and/or formulations, developed through a thorough analysis of the data. To avoid disease bias, it is important to design a prediction model for patients with breast cancer. We hypothesized that (1) body composition in breast cancer survivors is different from healthy individuals, (2) breast cancer disease and its treatment affect body composition, and (3) the currently used BIA algorithm may give erroneous results in survivors of breast cancer.

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MATERIALS AND METHODS

This research was a prospective validation study. The study was conducted between August 2019 and August 2020 in women with breast cancer who applied to Acıbadem Altunizade Hospital Department of Nutrition and Dietietic for a routine nutrition status assessment. As a standard method for comparison, DXA was used, and for the development of a predictive model, BIA was used, which is suitable for assessing the body composition of patients with breast cancer.

Patients and study design

The participants were patients whose treatment had been completed at least 6 months prior to the study. Height of participants was measured with SECA 264 digital height-measuring meter. BIA was performed under standardized conditions according to the manufacturer's protocol and scientific literature. All measurements were performed with light clothing and with metal objects (eg, jewelry, keys) removed. Before the body composition measurement, the necessary rules for the standardization of the measurement were communicated to the patients in written form and verbally.[13,14]

Individuals who participated in the study were asked not to undergo heavy physical activity 24–48 h beforehand, not to use alcohol 24 h beforehand, take the measurement while fasting, and not to have coffee or tea the night before the test.

Individuals were asked to remove clothing or jewelry that contained metal.[15] BIA and DXA measurements were made between 8:00 and 11:00 AM. DXA measurement was performed by a licensed technician in a temperature-controlled room. BIA measurement was performed by a dietitian in a temperature-controlled room. Before both measurements, participants rested for 10 min. After the BIA evaluation, a DXA measurement was performed.

Workgroup

Participants met the following inclusion criteria:

- Volunteers with a diagnosis of primary breast cancer who apply to the oncology diet Department of Nutrition and Dietietic for nutrition evaluation
- Women older than 18 years
- Those who have had lumpectomy and mastectomy
- Patients who completed neoadjuvant/adjuvant chemotherapy and/or radiotherapy treatment within the previous 6 months
- Patients with stages I–III according to the TNM (TNM classification of malignant tumors is a globally recognized standard for classifying the extent of spread of cancer)
- Patients with abnormalities in their body shape (eg, obesity, orthopedic prosthetics, implants, chronic diseases, and fluid disorders [edema]), to increase generalization

Exclusion criteria

Individuals without a cancer diagnosis, patients with metastasis, and patients with pacemakers and defibrillators were not included in the study.

Ethics

The study was found to be suitable for medical ethics by the Medical Research Evaluation Board (ATADEK) of Acıbadem Mehmet Ali Aydınlar University, and approval from the Ethics Committee was given (decision number 2019-13/12).

BIA evaluation

Tanita MC-980 Multi Frequency Segmental Body Analyzer was used for the BIA analysis of body composition. The impedance measurement system of the device was eight-electrode segmental multifrequency BIA, and the measurement frequencies were 1/5/50/250/500/1000 kHz. The device measures up to 300 kg, with an accuracy of 0.1 kg.

Before measuring body composition with the device, the individuals' heights were measured. Age, sex, and height measurements of each participant were recorded with standard mode in the Tanita computer program. Participants were asked to stand on the device such that their bare feet were in contact with the foot electrode pads. During the measurement, the electrode arms were held with both hands in a way that would allow the palms to grasp the arms, and participants were asked to move their arms away from their body at an angle of approximately 30°. After the patients were in the appropriate position, the measurement took 30 s. Body weight was measured by the system built into the device. Using a multifrequency impedance plethysmograph body composition analyzer, the device detected intracellular

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and extracellular water content separately. Data from the device included body composition measurements such as fat-free mass (FFM), FM, body fat percentage (%BF), and lean soft tissue mass (LSTM).

DXA evaluation

In the study, the DXA device branded "GE Lunar Prodigy Primo" was used for a full-body scan. Following the BIA measurements, participants' DXA measurements were performed by a licensed DXA technician. The DXA machine used in the study was calibrated daily. Participants were asked to remove clothing or accessories with metal during DXA measurement. Participants laid on the bed of the DXA scanner with their arms slightly separated from the torso. The entire body was scanned with the DXA device, starting from the head area and ending on the feet. According to the size of the body mass, the measurement was completed in 8–15 min. The DXA software then calculated the whole and regional body composition measure, including LSTM, FFM, FM, %BF, bone mineral content, and bone mineral density.

The classification described by Buckinx et al.[16] was used to define body compartments for the two devices.

Statistical analysis

Numerical variables were reported using mean, standard deviation, and minimum and maximum statistics, whereas categorical variables were reported using number and percentage. The level of linear correlation between BIA and DXA was assessed using Pearson correlation. The Bland-Altman plot was used to compare the consistency of the continuous data between the BIA and DXA clinical measurement methods. The Pearson correlation coefficient was used to identify the variable and predictive method with the best prediction.

Pearson correlation coefficient, Deming regression, and Bland-Altman analyses were performed to determine the compatibility between DXA and BIA

TABLE 1 Characteristics of participants ($n = 104$)

Variables	Data values
Age, mean ± SD (range), year	47.03 ± 8.59 (32–72)
BMI, mean ± SD	27.46 ± 5.16
Body weight, mean ± SD (range), kg	71.88 ± 13.41 (49.40–117.00)
Waist circumference, mean ± SD (range), cm	85.65 ± 11.49 (65.00–123.00)
Hip circumference, mean ± SD (range), cm	107.04 ± 10.05 (88.00–139.00)
Waist/hip ratio, mean ± SD (range)	0.80 ± 0.07 (0.67–1.14)
BMI classification, *n* (%)	
Normal	36 (34.6)
Overweight	43 (41.4)
Obese	25 (24.0)
Treatment, *n* (%)	
Surgical	104 (100)
Mastectomy	27 (26.0)
Breast protective surgery + SLNB	40 (38.5)
Breast protective surgery + ALND	37 (35.6)
Chemotherapy	84 (80.8)
Radiation therapy	82 (78.8)
Endocrine therapy	89 (85.6)
Menopause status, *n* (%)	
Premenopausal	84 (80.8)
Postmenopausal	20 (19.2)

Abbreviations: ALND, axillary lymph node dissection; BMI, body mass index; SLNB, sentinel lymph node biopsy.

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body composition values. Group comparisons were made with *t* test for two independent groups, whereas analysis of variance was used with more than two independent groups. Post hoc (least significant difference [LSD]) tests were performed when more than two group averages were found to have significant results after comparison. Statistical analysis was performed using SPSS (version 25) and MedCalc (version 15.8) software. The significance level was considered to be 0.05.

RESULTS

The characteristics of the participants (104 participants who survived nonmetastatic breast cancer) in the study are shown in Table 1. The average age of participants was 47.03 ± 8.59 years (range, 32–72), and mean BMI was 27.46 ± 5.16. According to BMI categories, 34.6% of participants were rated as having normal weight (18.50–24.99), 41.4% as overweight (25.00–29.99), and 24% as obese (>30.00). The mean body weight of the participants was 71.88 ± 13.41 kg (49.40–117.00), mean waist circumference was 85.65 ± 11.49 cm (65.00–123.00), and mean waist/hip ratio was 0.80 ± 0.07 (0.67–1.14). Twenty-six percent of the participants had a mastectomy, whereas the rest had breast-conserving surgery. When evaluated according to menopausal status, 80.8% of participants were diagnosed during the premenopausal period. Of the participants included in the study, 9.6% were assessed as premenopausal and others as postmenopausal (treatment-related, due to hysterectomy). The participants' mean time since cancer diagnosis was 41.82 ± 36.81 months (6–192 months). The mean change in body weight and BMI from time of diagnosis to study entry respectively was 4.03 ± 6.52 kg (−12.80 ± 18.9) and 1.47 ± 2.41 (−4.70 to 6.8).

When the participants were divided into BMI categories and examined, a statistically significant difference was found between BMI (normal, overweight, and obese) groups in both BIA and DXA measurements in terms of body components. A statistically significant difference was found between the mean of each group and the mean of the other group for each element ($P < 0.001$) (Table 2).

The distribution of body components for BIA and DXA measurements is shown in Table 3. A statistically significant difference was found between the measurement averages obtained from two different devices regarding the body components ($P < 0.001$). A statistically moderate relationship regarding the mineral measurements, a strong relationship regarding body FM (kg), and a strong relationship regarding other variables were found. The results indicate constant and/or proportional error, as well as a statistically moderate relationship between the two devices, especially in terms of strong and very strong components.

TABLE 2 Comparison of body composition according to BMI groups

			BIA				DXA			
Body composition	**BMI**	***n***	**Mean**	**SD**	***F***	***P***	**Mean**	**SD**	***F***	***P***
FFM, kg	Normal	36	42.43[a]	3.21	59.142	<0.001	37.77[a]	3.07	35.551	<0.001
	Overweight	43	46.21[b]	3.20			40.68[b]	3.65		
	Obese	25	53.14[c]	5.25			46.42[c]	5.34		
FM, kg	Normal	36	17.52[a]	4.11	136.749	<0.001	23.09[a]	3.74	160.576	<0.001
	Overweight	43	24.92[b]	3.76			30.65[b]	3.40		
	Obese	25	37.46[c]	6.42			43.99[c]	6.64		
%BF	Normal	36	28.97[a]	5.25	61.679	<0.001	39.30[a]	4.33	53.955	<0.001
	Overweight	43	34.70[b]	3.83			44.33[b]	3.66		
	Obese	25	41.17[c]	3.09			49.85[c]	3.71		
LSTM, kg	Normal	36	40.26[a]	3.05	43.677	<0.001	35.42[a]	2.86	38.893	<0.001
	Overweight	43	44.40[b]	4.48			38.45[b]	3.54		
	Obese	25	50,45[c]	5,01			44,07[c]	5,12		

Note: [a,b,c]Pairwise comparison is summarized with high indices. Analysis of variance test (least significant difference post hoc testing done for pairwise comparison).

Abbreviations: BIA, bioelectrical impedance analysis; BMI, body mass index; DXA, dual-energy x-ray absorptiometry; FM, fat mass; FFM, fat-free mass; LSTM, lean soft tissue mass; %BF, body fat percentage.

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TABLE 3 Comparison of BIA and DXA body composition measurements

	BIA	DXA	Difference	t[a]	P	r[b]	CCC
FFM, mean ± SD, kg	46.57 ± 5.54	41.06 ± 5.11	5.51	22.173	<0.001	0.890	0.576
FM, mean ± SD, kg	25.37 ± 8.84	31.24 ± 9.09	−5.87	−26.280	<0.001	0.968	0.796
%BF, mean ± SD	34.28 ± 6.24	43.91 ± 5.58	−9.64	−31.304	<0.001	0.865	0.368
LSTM, mean ± SD, kg	44.42 ± 5.66	38.75 ± 4.98	5.67	18.705	<0.001	0.839	0.530

Abbreviations: BIA, bioelectrical impedance analysis; CCC, concordance correlation coefficient; DXA, dual-energy x-ray absorptiometry; FM, fat mass; FFM, fat-free mass; LSTM, lean soft tissue mass; %BF, body fat percentage.

[a]Paired t test for two dependent groups.

[b]Pearson correlation coefficient.

Accordingly, as seen in the Deming regression and Bland-Altman graphs, BIA results for FFM (Figure 1) included high measurement values with a difference of 12.6% when compared with DXA. The following mathematical relation has corrected the resulting fixed proportional error:

$$FFM\,(\text{kg})DXA = \frac{FFM\,(\text{kg})(BIA) - 2.0470}{1.0844}$$

Two BIA devices were used to measure FM (kg) (Figure 2), and compared with DXA, the BIA results reported lower measurement values, with an average difference of 5.9 kg (Figure 2). A statistically significant, fixed error was shown between measurements. The proportional fixed error between the BIA results and the DXA results can be corrected by using the following equation:

$$FM\,(\text{kg})DXA = \frac{FM\,(\text{kg})(BIA) + 5.0319}{0.9731}$$

Compared with DXA, BIA results reported lower % BF by a difference of 9.6% (Figure 3). The difference in measurement between the two devices was higher at low weights. The following equation was found to fix the BIA result according to the DXA result, taking into account that the margin of error is high in thin individuals:

$$BF\,(\%)DXA = \frac{BF\,(\%)(BIA) + 14.8930}{1.1197}$$

For LSTM measurement (Figure 4), BIA results were found to measure LSTM higher than DXA, with an average difference of 13.6%. There are proportional deciduous errors between measurements. The following equation has been found for adapting the BIA value to the DXA data:

$$LSTM\,(\text{kg})DXA = \frac{LSTM\,(\text{kg})(BIA) - 0.3479}{1.1373}$$

As a result of the study, the above-mentioned mathematical equations were derived, which allow us to achieve DXA results with the data obtained from BIA. A mathematical correlation was not given for the percentage of fat in the legs, percentage of fat in the body, and mineral mass because they had no decisive correlation between each other.

DISCUSSION

To our knowledge, this is the first study in the literature to validate the BIA method and the mathematical relationship for correction in evaluating body composition in patients with breast cancer.

The importance of body composition for patients with breast cancer is slightly different than for those with other cancers: (1) adjuvant endocrine therapy applied may have an effect on body composition that increases FM, and (2) obesity and sarcopenic obesity are independent factors that negatively affect the prognosis.[2–6,17–19] BIA reports are based on formulating measurements from healthy individuals, and body compositions of patients with breast cancer may different from those of healthy individuals.

The only study similar to ours in the literature is by Lee et al.,[20] who compared the measurement methods of BIA and DXA for the evaluation of body composition in patients with breast cancer. Their study, which included 89 patients with nonmetastatic breast cancer, divided individuals into four groups according to their BMI as "normal," "overweight," "obese," and "morbidly obese," comparing the differences between the two devices. The %BF was 44.2% ± 6.2% for DXA and 40.4% ± 7.8% for BIA; lean body mass was 39.1 ± 7.6 kg for DXA and 42.9 ± 5.9 kg for BIA; and FM was 32.4 ± 10.8 kg for DXA

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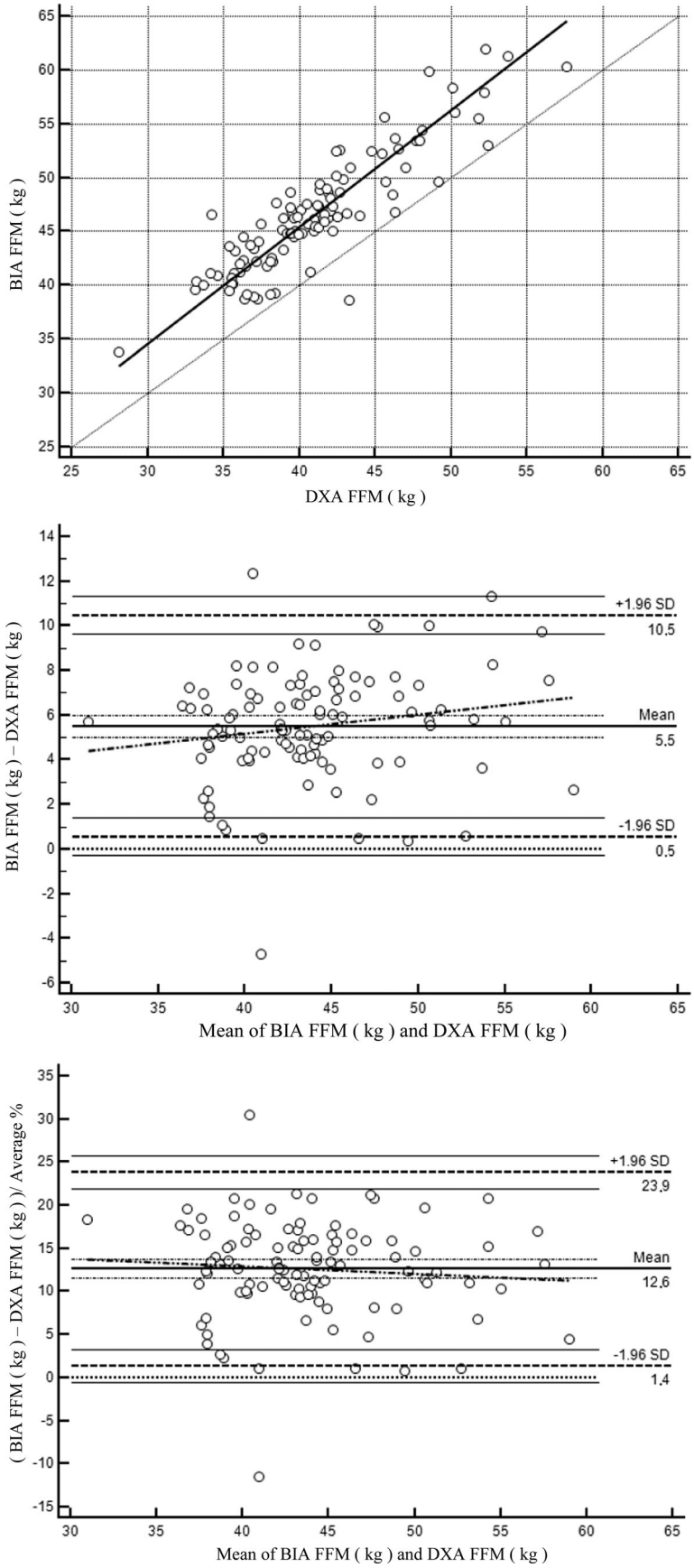

FIGURE 1 FFM regression curve and Bland-Altman analysis. BIA, bioelectrical impedance analysis; DXA, dual-energy x-ray absorptiometry; FFM, fat-free mass

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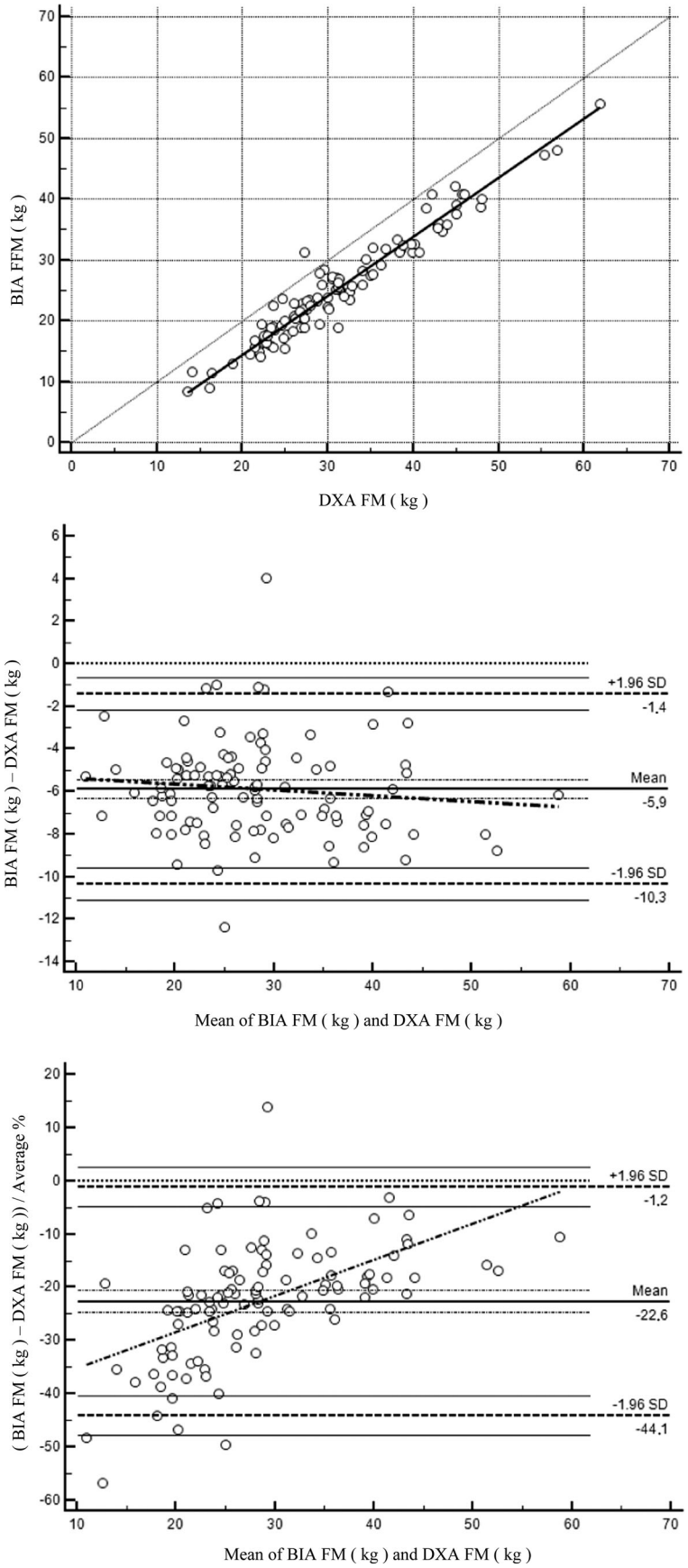

FIGURE 2 FM correlation curve and Bland-Altman analysis. BIA, bioelectrical impedance analysis; DXA, dual-energy x-ray absorptiometry; FM, fat mass

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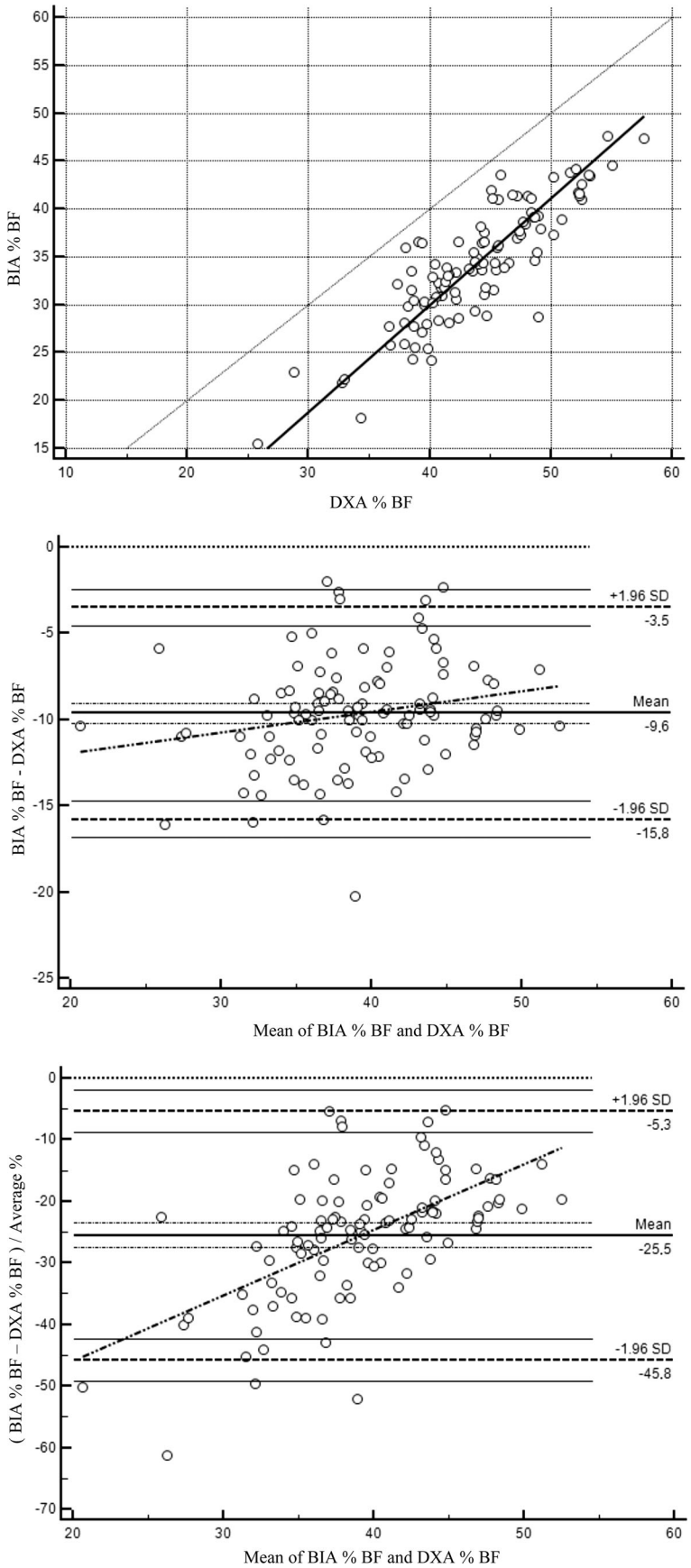

FIGURE 3 %BF correlation curve and Bland-Altman analysis. BIA, bioelectrical impedance analysis; DXA, dual-energy x-ray absorptiometry; %BF, body fat percentage

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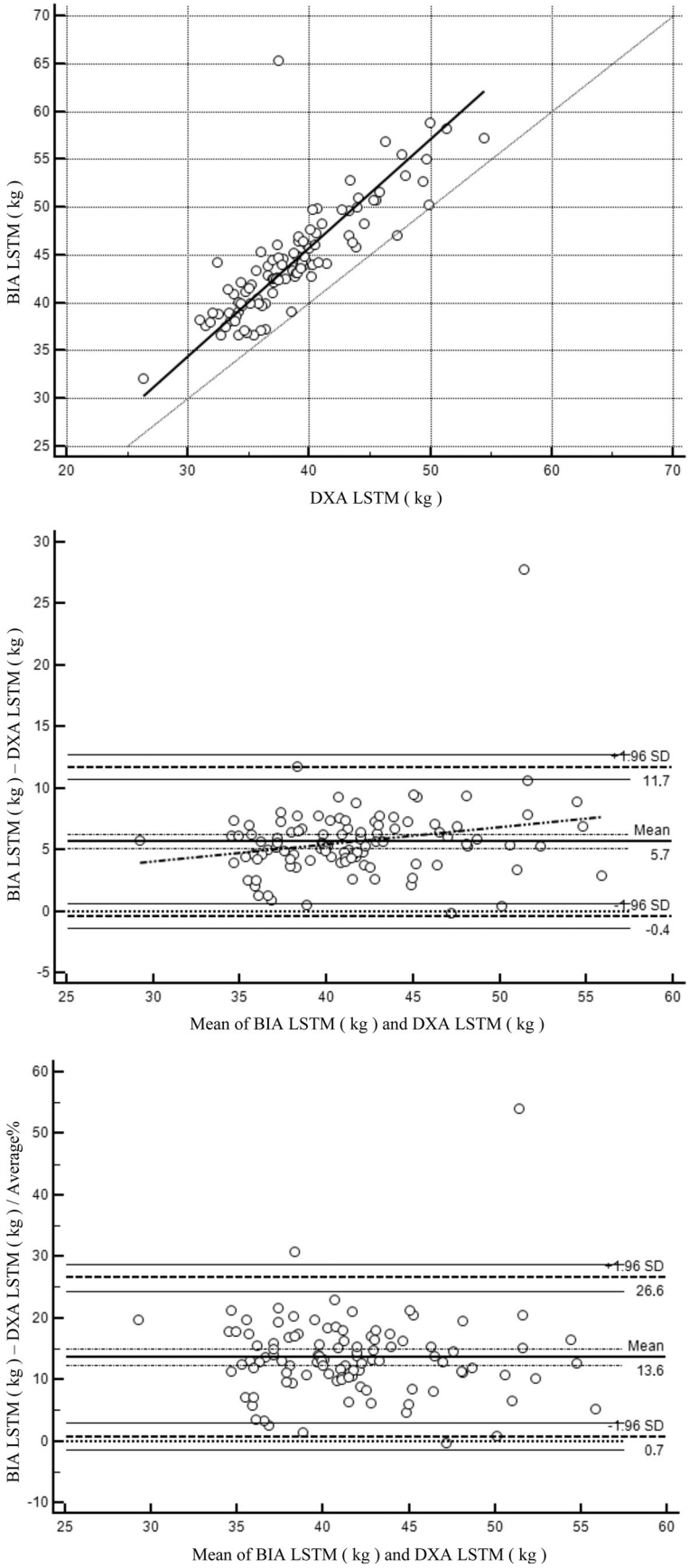

FIGURE 4 LSTM correlation curve and Bland-Altman analysis. BIA, bioelectrical impedance analysis; DXA, dual-energy x-ray absorptiometry; LSTM, lean soft tissue mass

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and 30.6 ± 11.0 kg for BIA ($P = 0.001$). It was reported that there was significantly different body compositions based on the two devices in the normal, overweight, and obese groups of patients with breast cancer. By contrast, in the morbidly obese group ($P = 0.102$), there was consistency for FM between the two devices (DXA: 48.7 ± 7.2 kg vs BIA: 47.9 ± 5.7 kg), which probably results from the small sample size. The authors concluded that BIA estimates %BF and FM lower than the actual values and estimates FFM higher than DXA estimates in patients with breast cancer. The authors proposed that studies with broader BMI categories should be conducted.

Our results are in line with the results of Lee and colleagues in terms of FFM, %BF, and FM. In our study cohort, there were 25 participants in the obese category; 7 participants had a BMI of >35, and 3 participants had a BMI of >40. According to the classification by Lee and colleagues, in our study cohort, the rate of participants with morbid obesity was 9.6%.

Zhang et al.[11] showed an association between sarcopenia and mortality in female patients with breast cancer. The findings showed that patients with breast cancer and sarcopenia had a 71% increased risk of mortality compared with patients without sarcopenia but there was high heterogeneity. Furthermore, they conducted a subgroup analysis according to disease stages and found that the presence of sarcopenia in nonmetastatic breast cancer increased the risk of mortality.[11] Thus, evaluating body composition in patients with breast cancer remains a critical issue in the disease progression. The Health, Eating, Activity, and Lifestyle (HEAL) Study cohort of breast cancer survivors reported that sarcopenia might be an independent predictor of mortality regardless of obesity.[21] Additionally, based on the results of this study, it was reported that there is a need to examine the potential use of body composition assessment and validate the findings.[21] The SCAN study was a cross-sectional multicenter French study that aimed to estimate sarcopenia prevalence in a real-life sample of patients with metastatic cancer. This study evaluated 139 patients with breast cancer who had a median age of 61.2 years (29.9–97.8 years) and showed that 29.5% had sarcopenia, whereas 41.0% had presarcopenia.[22]

DXA is a reliable measurement method for measuring body composition because of its high validity and reproducible properties.[23] However, DXA use is limited owing to a lack of availability and high cost. By contrast, BIA is a more commonly used tool for predicting body composition because of its safety, accuracy, low cost, and ease of use.[24] BIA has been widely used in breast cancer studies and as part of clinical examination. Still, little is known about whether the BIA results obtained when evaluating body composition of patients with breast cancer are compatible with the results of DXA measurements.

In a study by Völgyi et al., the body composition of 82 males and 86 females between 37 and 81 years old were evaluated using DXA and two different BIA devices. Subjects were divided into two groups as having low or high levels of physical activity. According to their BMI, they were categorized as normal, overweight, or obese.[25] In a study of 551 elderly individuals, BIA (InBody) overestimated FM and underestimated muscle mass.[26] However, a study conducted in Finland of 882 adults between 18 and 88 years, showed that full-body muscle mass was overestimated based on (InBody 720).[27] In their study of a healthy population in Taiwan, Chen et al.[28] investigated differences in sex and degrees of obesity to confirm the results of the BIA (Tanita BC-418) measurement. The authors reported that BIA measurement results became increasingly inaccurate as obesity rating increased.[28] Another study reported that the percentage of fat in women estimated by BIA was incomplete, and the deviation increased as the percentage of fat increased.[29] Sun et al. reported that BIA (Bodystat QuadScan 4000) tends to predict a low fat ratio as obesity rates decrease. By contrast, BIA tends to indicate a high fat ratio in thin individuals. The degree of obesity of an individual is recognized as an essential factor in affecting the accuracy of measurement.[30] In a study that monitored longitudinal change in FFM in patients with colorectal cancer and validated BIA, it was reported that when compared with DXA, BIA overcalculated FFM for both loss and gain but showed 100% sensitivity and 90% specificity in FFM losses which were >5%. The authors reported that more studies are needed to validate BIA in tracking changes in lean body mass.[31] In a study that validated BIA in the evaluation of body composition of patients with colorectal cancer, it was reported that BIA is a valuable tool for identifying low fat-free body mass as part of the diagnosis of malnutrition of BIA.[32] One validation study was conducted in a small sample ($n = 17$) of patients with chronic obstructive pulmonary disease. BIA data were reported to be compatible with DXA.[33] It is essential to evaluate body composition and sarcopenic obesity with valid measurement tools in patients with breast cancer. Validation of BIA measurement, a noninvasive and inexpensive method, in patients with breast cancer may allow this tool to be used more widely in patients with breast cancer.

According to Ward and Müller,[34] BIA is a central tool in the clinical examination of patients to address hydration, functional body mass, energy stores, and prognosis just as one would consider an electrocardiogram for the interpretation of cardiac function.

However, a recent article by Ward[35] reported that three key issues with regards to BIA:

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- Misunderstanding of, or an unwillingness to accept, the limitations of the impedance method and a desire for performance beyond these limits
- Distinction between statistically significant differences and clinically relevant differences
- Lack of standardization of impedance protocols

Patients with breast cancer may be referred to Department of Nutrition and Dietetics for treatment to lower their body fat to healthy limits, lose weight, and diagnose insulin resistance. In these patients, in whom body composition change is critical in the follow-up and treatment process of their cancer, correcting the proportional error caused by BIA has a vital role in determining lifestyle change and nutrition strategies. The mathematical equality we present can be safely used not only because there is a proportional constant error between the two devices for FFM, FM, %BF, and LSTM, but also because it provides a strong correlation.

The limitation of this study is that although patients with metastasis are excluded, breast cancer contains a heterogeneous group of histological subtypes. In addition, our results only determined the correction approach of the predictive algorithms used in the Tanita MC-980 impedance device. It would be appropriate to perform similar studies for other devices. Also, an estimation algorithm improved by raw impedance data might be a better method. Thus, it will be possible to develop correction formulas regardless of the device brand. The efforts regarding the shortening of patients' time in the hospital and the limitation of accepting new patients due to the COVID-19 pandemic reduced the number of available patients.

CONCLUSION

Oncology patients undergo a dynamic treatment process, and their nutrition status often may vary. Today, breast cancer is among the cancers with the highest survival rate. The number of patients in the follow-up process increases exponentially with increasing diagnostic tools and life expectancy. Studies of lifestyle changes are gaining importance to improve the quality of life and survival rates of patients with breast cancer. Body composition evaluation is often involved in the scope and focus of these studies in question. The validation of BIA, which is a noninvasive, easy, inexpensive, and frequently repeatable measurement method regarding the evaluation of body composition, can bring convenience and standards for body composition assessment in future studies. By the mathematical relationship between the two measurement methods, it seems possible to adapt the body composition parameters obtained from the BIA of patients with breast cancer to DXA results. BIA results can be verified using the mathematical equations given in the research results.

Both body fat and LSTM are parameters that should be carefully monitored in survivors of breast cancer.

Closely monitoring patients' body compositions together with adjuvant endocrine treatment, correcting BIA results with the formulas we found from our research, and individually planning and monitoring nutrition and physical activity recommendations according to verified results may increase follow-up success.

CONFLICT OF INTEREST

None declared.

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